Assignment 2

CS 2470 Questions

Question 1

Proof: Let us proof the transnational equivariance for 1D vector.
We want to proof that $F(T(x)) = T(F(x))$, where $F(x), T(x)$ are convolution and translation, respectively. Consider an input vector $\{x|x \in R^n, x_i = 0, i \in [1, n] \setminus \{k, k+1, ..., k+p\}\}$. A filter $\{w|w \in R^m\}$ has a smaller size than the input, namely $n > m$.

$$F(T(x)) = \sum_{i=1}^{n-m+1} \sum_{j=1}^{p} w_j * T(x_{i+j}), \tag{1}$$

$$= \sum_{i=k-m+1}^{k+m-1} \sum_{j=1}^{p} w_j * x_{i+j}, \tag{2}$$

$$T(F(x)) = T(\sum_{i=1}^{n-m+1} \sum_{j=1}^{p} w_j * x_{i+j}), \tag{3}$$

$$= \sum_{i=k-m+1}^{k+m-1} \sum_{j=1}^{p} w_j * x_{i+j}, \tag{4}$$

Hence convolution is equivariant under translation.

Question 2

(a) There should be (2*2*3)*3's (36) learnable weight parameters.
(b) The output shape is [27, 27, 3], and the input tensor shape is [28, 28, 3]. To replicate such operation in a fully-connected layer, the total number of parameter should be 27*27*3*28*28*3 + 27*27*3 = 5146011